APPLIED PHYSICS LABORATORY MANUAL

(AS PER 2025 SYLLABUS)

FOR B. TECH

NAME OF THE STUDENT: ____________________

SECTION AND BRANCH: ____________________

ROLL NO. OR U S N : ____________________

DEPARTMENT OF PHYSICS,
FACULTY OF ENGINEERING AND TECHNOLOGY
JAKKASANDRA POST, KANAKAPURA TALUK, BANGALORE SOUTH DIST.
562 112

LABORATORY INSTRUCTIONS

1. **Required Materials:**
   Students must bring the laboratory manual, laptop, calculator, and other required materials for each practical class.
2. **Preparation:**
   Students are expected to come well-prepared to conduct the scheduled experiment.
3. **Recording and Verification:**
   All measurements, readings, and calculations must be recorded, verified, and acknowledged by the concerned faculty member.
4. **Tools for Calculations:**
   Calculations must be performed using appropriate tools such as Microsoft Excel or Python.
5. **Submission of Records:**
   Students must submit the completed laboratory manual along with the data sheet in the following week for review and correction.
6. **Evaluation and Internal Assessment:**
   Failure to obtain and get the results evaluated during the same laboratory session will result in the loss of Internal Assessment marks for that experiment.
7. **Care of Equipment:**
   Students are responsible for the proper handling and safe usage of the apparatus provided.
8. **End-Semester Practical Examination:**
   A laboratory test will be conducted at the end of the semester.
9. **Discipline:**
   Strict discipline must be maintained in the laboratory at all times.

VISION

We are committed to enhance the linkages between engineering and other professions with the aim of tackling increasing complex challenges that lie at the interfaces of science, technology, and society and to dynamically integrate the components of sciences, humanities and Engineering to groom the students to become globally acknowledged professionals and to continually improve the research skills of the students.

MISSION

- Advancing the frontiers of knowledge in the physical, chemical and mathematical sciences.
- To widen intellectual perspective and enhance entrepreneurship qualities.
- Develop technical and communicative skills to make the students industry ready.
- To inculcate ethical values and promote social responsibility through an innovative learning process in basic sciences and humanities.
- To provide students with soft skills and behavioral training programs in order to develop their overall personality and social consciousness.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

- To be successful in professional career by acquiring the knowledge in the fundamentals of Physics principles and professional skills.
- To be in a position to analyze real life problems and design socially accepted and economically feasible solutions in the respective fields.
- To involve themselves in lifelong learning professional development by pursuing higher education and participation in research and development activities.
- To exhibit good communication skills in their professional career, lead a team with good leadership traits and interpersonal relationship with the members related to engineering streams

Applied Physics Laboratory

Subject Code: **Credits:4**

Hours/week: 2

COURSE LEARNING OBJECTIVES:

The aim of this course is to:

- Develop a comprehensive understanding of fundamental physics concepts in quantum mechanics, photonics, oscillations, and semiconductor physics while integrating computational problem-solving techniques.
- Master modern physics applications including quantum computing algorithms, laser systems, optical communication technologies, and semiconductor devices.
- Cultivate systematic experimental design, data analysis, and scientific documentation capabilities through hands-on laboratory work and computational simulations.

Sl No	List of Experiments
1.	**Planck's constant-** Determination of the Planck's constant using light emitting diodes.
2.	**Laser diffraction:** Determination of wavelength of given laser.
3	**Band gap-** Determination of Energy gap of a given semiconductor.
4.	**Torsional Pendulum-** Determination of moment of inertia of the given irregular body.
5	**Quantum Gates**- Construction of Quantum gates using Infosys or IBM portals Simulation of Single-Qubit Gates using Qniverse/Quirk

Course Outcomes:

Sl. No.	Course Outcome	Description	Bloom's Taxon Level
1.	CO1	**Apply** quantum mechanical principles using structured computational frameworks to solve physics and engineering problems, evaluating theoretical predictions against computational results.	Applying (3)
2.	CO2	**Analyze** quantum computing concepts and algorithms for information processing applications, demonstrating proficiency with quantum programming platforms and advanced computational technologies.	Analize (4)
3.	CO3	**Demonstrate** photonic principles in optical communication systems using computational simulations, analyzing light-matter interactions for modern engineering applications.	Understanding (2)
4.	CO4	**Simulate** oscillatory systems and wave phenomena using advanced computational techniques, critically evaluating resonance behavior through structured mathematical and programming approaches.	Applying (3)
5.	CO5	**Evaluate** semiconductor materials and electronic devices using computational modelling and band theory principles, applying programming tools to predict material behavior in engineering contexts.	Evaluate (5)
6.	CO6	**Execute** structured computational experiments and systematic data analysis protocols, documenting scientific observations using standard technical formats and presenting findings through programming-based visualizations.	Applying (3)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	3	2	2	2	-	-	-	1	-	-
CO2	3	3	2	2	2	-	-	1	1	-	-
CO3	3	2	3	2	2	1	-	-	1	-	-
CO4	3	3	2	2	2	-	-	-	1	-	
CO5	3	3	2	2	2	-	-	-	1	-	-
CO6	2	2	-	3	3	-	-	2	3	-	-
Avg	2.83	2.67	2.2	2.17	2.17	1	-	1.5	1.33	-	-

INDEX SHEET

Sl. No.	Name of the Experiment	Page number	Conduction date	Marks obtained	Signature (Staff)
1	Planck's constant	7			
2	Laser diffraction	11			
3	Band gap of semiconductor	15			
4	Torsional pendulum	19			
5	Quantum Gates	25			
Average marks					

Date:

Signature

(Staff Incharge)

1. Planck's Constant

Aim: To determine the Planck's constant using light emitting diodes.

Apparatus: Light Emitting Diodes of 5 different wavelengths, power supply and multimeter.

Formula: $\boldsymbol{E = h\,\nu\ J}$

$$\boldsymbol{h = \frac{e(V_k\lambda)_{mean}}{c}\ Js}$$

Here,

E is the energy of the photon, J

$\boldsymbol{h}$ is Planck's constant, Js

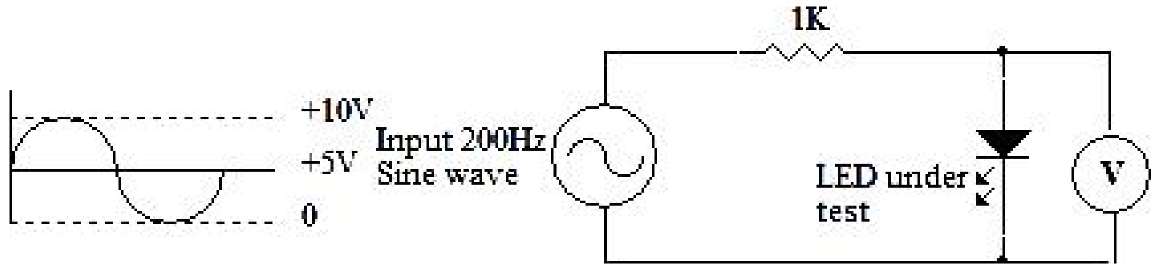
$\boldsymbol{\nu}$ is the frequency of the radiation, Hz

Procedure: Circuit connections are made as shown in the circuit diagram. Power supply is switched on and using a digital peak reading voltmeter the voltage (Vk) across the LED is measured and recorded in table for given color LED light. This is repeated for the other four LEDs. Energy of the light radiation is calculated using the equation E= e Vk. Here 'e' is the charge on electron 1.6 x 10^{-19}C. The frequency of the light radiation is determined using $\boldsymbol{\nu = c/\lambda}$. Here 'C' is the velocity of light (3 x 10^{8} ms^{-1}) and '$\boldsymbol{\lambda}$' is the wavelength of light emitted. A plot of energy against frequency is made. According to Planck's Quantum theory the energy and frequency relationship for the radiation is given by $\boldsymbol{E = h\nu}$. Here '$\boldsymbol{h}$' is Planck's constant. Thus, the slope of the curve gives the Planck's constant.

[Note: LED is P-N junction made of heavily doped transparent semiconductor. When it is forward biased, if the applied voltage is higher than the knee voltage then electrons and holes from N and P sections recombine in the depletion region resulting in the emission of photons. Thus, LED glows with characteristic wavelength which depends on the composition and condition of the semiconductor material used. When the applied voltage is equal to the turn on voltage the LED just glows, and the energy of the photons emitted is equal to the energy acquired by the electron from the electric field. Thus, energy of the photon can be calculated from the turn on voltage knowing the wavelength of the emitted radiation a plot of energy versus frequency can be made. Thus, the Planck's constant can be determined.**]**

Observation:

Circuit diagram

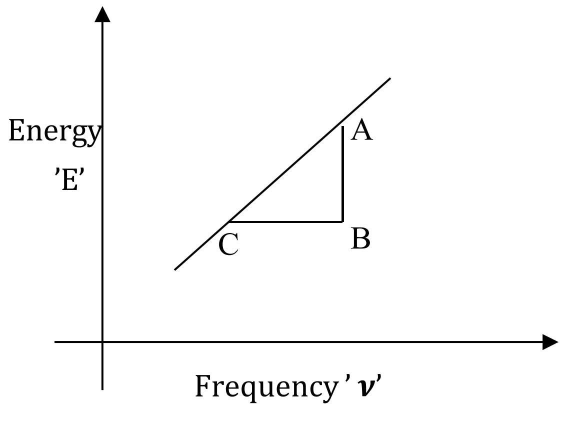


LED	Color	Wavelength (λ in m)	Knee Voltage (V_k in Volts)	Energy of Radiation $E = e\,V_k$ (J)	Frequency of the Radiation $\nu = c/\lambda$ (Hz)	$V_k\lambda$
1	Blue	346 X 10^{-9}				
2	Green	538 X 10^{-9}				
3	Yellow	568 X 10^{-9}				
4	Red	630 X 10^{-9}				
5	IR	940 X 10^{-9}				

$(V_k\lambda)_{mean} =$

$$h = \frac{e(Vk\lambda)\text{mean}}{c} \quad \text{Js}$$

h=..............Js



The Slope of the Curve = Planck's constant = $h = \frac{AB}{BC}$ =____________________Js

Result: The Planck 's Constant is given by

By graph, h=_______________Js

By calculation, h=__________________Js

2. LASER DIFFRACTION

Aim: To Determination wavelength of given laser light

Apparatus: Diffraction grating, Laser source, grating stand, screen.

Formula: $\lambda = \frac{2C \sin\frac{\theta}{2}}{n}$ m

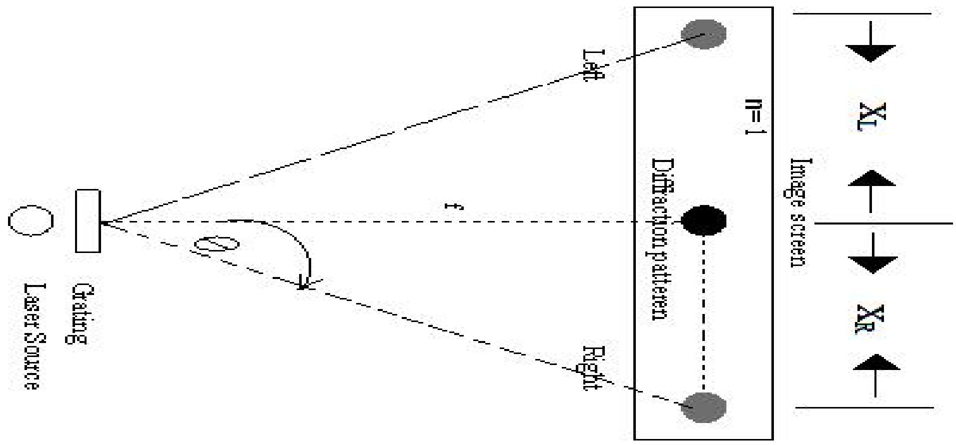
Where ,

- C, is grating constant (m)
- θ, is the angle of diffraction(degree)
- n, is order of diffraction
- λ, is the wavelength of the laser light used (m)
- The angle of diffraction θ is given be $\theta = \tan^{-1}\frac{X}{f}$
- x, is central maxima and secondary maxima.
- f , is the distance between the image and the grating.

Procedure:

The laser source is placed on a table and switched on. The leveling screws of the laser are adjusted such that the laser beam exactly falls on center of graph sheet placed on the screen, which is placed at about 1m from the laser source. The diffraction grating is now placed on grating stand close to the laser source. The diffracted laser spots are observed on either side of central maximum. The central maximum is very bright and as the order of diffraction increases the brightness decreases. The center of various spots of the diffraction pattern are marked using a pencil, then the graph sheet is removed from the screen and the distances between central maxima and various diffracted spots are measured on either side of central maximum. Diffraction angles are calculated for various orders of diffraction. The wavelengths of laser for various orders of diffraction are calculated using given formula and the average wavelength is obtained.

Observations:



Tabular column:

Sl. No.	Order of spectrum (n)	Distance from central maximum (cm)		Mean distance Xn = (XL+ XR)/ 2 (cm)	$\theta = \tan^{-1}\frac{X}{f}$ (deg)	λ (m)
		LHS (X_L)	RHS (X_R)			
1	1					
2	2					
3	3					
4	4					
5	5					
6	6					
7	7					
8	8					

Mean λ =…………(m)

Calculations:

Distance between the grating and screen, f = 80 cm

The angle of diffraction, $\theta = \tan^{-1}\frac{X}{f}$, θ =…… ….. (deg)

The distance between two consecutive rulings on grating K = 500 LPI

No of lines per meter on grating, N=$\frac{K}{2.54\times10^{-2}}$

Grating Constant, $C = \frac{1}{N}$ $in\ m$

C=……………..m

The wavelength of given laser source, **λ** =$\frac{2C\sin\frac{\theta}{2}}{n}$ m

$\boldsymbol{\lambda}$ = ____ m

λ = ____________m

Result: Wavelength of the given laser source is λ =_________________m

3. Band gap of a Semiconductor

AIM: To determine the energy gap of the given semiconductor.

APPARATUS: Semiconductor diode, thermometer, liquid paraffin, beaker, test tube.

FORMULA:

The band gap of the given semiconductor is

$$E_G = e\, Y_{\mathrm{int}ercept}$$

Where,

- 'e' is electronic charge of electron.
- $e = 1.6 \times 10^{-19}\ C$
- '$Y_{intercept}$' is intercept in the Y-axis by extrapolated straight in the graph.

Procedure:

1) Take water in a beaker.

2) The beaker is then heated by means of a stove. While the heating is in progress, the following connections are made.

3) The milli ammeter is connected between A and B, the voltmeter is connected between C and D as shown in circuit diagram.

4) The p-side lead of the semiconductor is connected to E and the n-side lead is connected to F.

5) The mercury bulb of the thermometer is tied to the semiconductor and it is inserted carefully into a test tube.

6) The test tube is then slowly placed into the water bath.

7) Once the temperature reading in the thermometer becomes 70 ^{O}C, the heating is stopped.

8) Note down the forward voltage across the semiconductor as read by the voltmeter starting from 70 ^{O}C to 50 ^{O}C in steps of 5 ^{O}C.

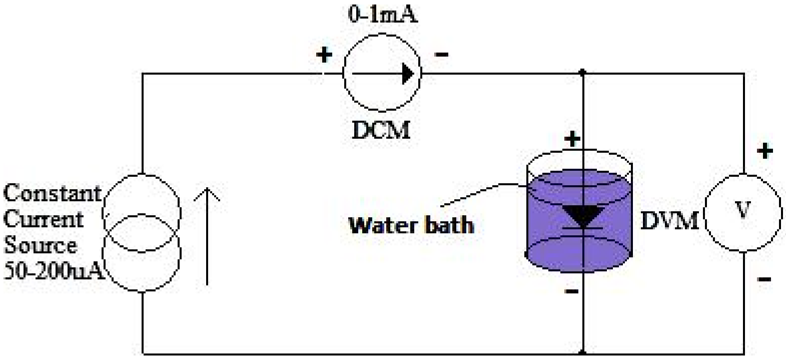
9) The forward current IF is maintained constantly at 0.1 mA by adjusting the knob, every time the readings are recorded.

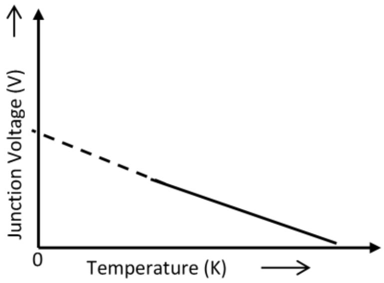
10) A graph is plotted by taking temperature in absolute degrees along X-axis and junction voltage along Y-axis as shown in graph.

11) A straight line is obtained which is extrapolated to cut the Y-axis (Y intercept). Then the Eg in eV is calculated using given formula.

Observations:

<u>Circuit diagram</u>





To determine the dependence of Junction Voltage on Temperature:

Forward constant current IF = 0.1 mA

Sl. No.	Temperature (°C)	Temperature T (K)	Junction Voltage (V)
1			
2			
3			
4			
5			
6	Room Temp		

Calculations:

The intercept value on Y-axis, $Y_{intercept}$ =.........Volts.

The band gap of the given semiconductor is

$E_G = e\, Y_{intercept}$

=.............. J

=............. eV

Result:

The band gap of the given semiconductor is found to be, Eg =................................eV.

4.T ORSIONAL PENDULUM

Aim: To determine the moment of inertia of an irregular body and also to determine the rigidity modulus of the material of the wire by setting up a torsional pendulum.

Apparatus: Rectangular plate, circular disc, irregular plate, stands with clamp, stop clock and experimental material wire.

Formula: $$I_0 = \left(\frac{I}{T^2}\right)_{mean} {T_0}^2 \qquad kgm^2$$

$$\eta = \frac{8\pi l}{r^4}\left(\frac{I}{T^2}\right)_{mean} \qquad Nm^{-2}$$

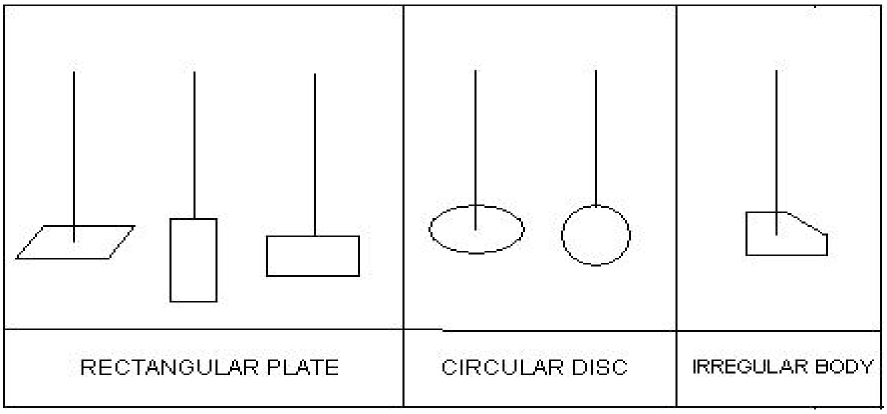
where,

- I_0 is the Moment of inertia of irregular body (kgm^2).
- T_0 is the time period of oscillation of the irregular body about a given axis (sec).
- I is the Moment of inertia of regular body on a given axis (sec).
- η is the Rigidity modulus of the material of the suspension wire (Nm^{-2}).
- l is the length of the wire between the chuck nuts (m).
- r is the Radius of the suspension wire (m).

Procedure: The masses of the given rectangular plate and circular plate are determined using weighing balance. The length L, breadth B of the rectangular plate and radius R of the circular plate are measured. The moment of inertia of these two bodies are calculated for different axes using the equations given in the table. The rectangular plate is suspended with the help of chuck nuts to the experimental wire with its axis perpendicular to the plane of the rectangular plate. Then the plate is set into torsional oscillations and its period for 10 oscillations is calculated. For the same rectangular plate, the time period is obtained in two other axes viz, an axis perpendicular to the length and an axis perpendicular to breadth. Then the experiment is repeated for circular disc with definite axis viz, perpendicular to its plane and along the diameter. Then the time period is noted for 10 oscillations. Then mean value of (I/T^2) is calculated.

The time period is noted for 10 oscillations for the irregular body whose moment of inertia about an axis is calculated using the given formula. The diameter of the wire is measured by using screw gauge and length of the wire between chuck nuts is measured using the scale. The rigidity modulus (η) of the wire is calculated using the given formula.

Observations: <u>Diagram</u>



Mass of the circular plate M =....................Kg

Radius of the circular plate R = (Circumference/ 2π) =.................... m

Mass of the rectangular plate M =....................Kg

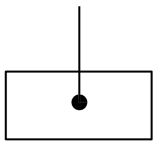
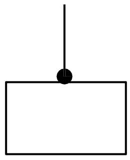
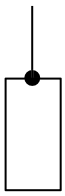
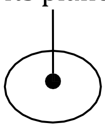
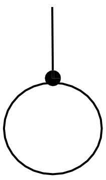
Length of the rectangular plate L =..................... m

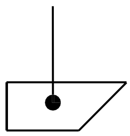
Breadth of the rectangular plate B =........................m

Length of the wire between the chuck nuts

Radius of the given wire r =................ m

Table: To find the time period (T)

Body	Axis	Moment of Inertia (*I*) Kg-m^2	Time taken for 10 oscillations 't' sec			Period $T = t/10$ sec	T^2	$\frac{I}{T^2}$ Kg-m^2/s^2
			t_1	t_2	$\frac{(t_1 + t_2)}{2}$			
Rectangular plate	Perpendicular to its plane 	$I_1 = \frac{M(L^2+B^2)}{12}$ $I_1 =$ ____						
	Perpendicular to its length 	$I_2 = \frac{ML^2}{12}$ $I_2 =$ ____						
	Perpendicular to its breadth 	$I_3 = \frac{MB^2}{12}$ $I_3 =$ ____						
Circular plate	Perpendicular to its plane 	$I_4 = \frac{MR^2}{2}$ $I_4 =$ ____						
	Along the diameter 	$I_5 = \frac{MR^2}{4}$ $I_5 =$ ____						
						$\left(\frac{I}{T^2}\right)_{Mean} =$ ________		

Body	Axis	Time taken for 10 oscillations 't' sec			Period $T_0 = t/10$ Sec
		t_1	t_2	$\frac{(t_1 + t_2)}{2}$	
Irregular body	Perpendicular to its plane 				

Calculations:

1. Moment of Inertia of the irregular body (Perpendicular to the plane)

$$I_0 = \left(\frac{I}{T^2}\right)_{mean} {T_0}^2 \quad kgm^2$$

$$I_0 = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \; kgm^2$$

2... Rigidity modulus of the given material wire

$$\eta = \frac{8\pi l}{r^4} \left(\frac{I}{T^2}\right)_{mean} \quad Nm^{-2}$$

$$\eta = \ldots\ldots\ldots\ldots\ldots\ldots \; Nm^{-2}$$

Result:

The moment of inertia of an irregular body is, I_o =.......................... Kgm^2

The rigidity modulus of the given wire is, η = ...Nm^{-2}

5. Quantum Gates

Aim

To simulate and analyze the behavior of single-qubit gates using online simulator, and to visualize their action using Bloch sphere schematics and amplitude/probability descriptions.

Apparatus Required

• Computer or laptop with internet access

• Web browser (Google/microsoft edge/ safari)

Simulator weblink:

https://algassert.com/quirk#circuit={%22cols%22:[],%22init%22:[%22-%22]}

Procedure:

1. Open the simulator in a browser and create a new single Qubit circuit.
2. Select the X quantum gate and give the $|\mathbf{0}\rangle$ input to the gate and observe the output through Bloch sphere.
3. Validate the output in the table (True or False).
4. Measure the angle **θ** and **ϕ** record in the table.
5. Similarly repeat the procedure for other input conditions.
6. Repeat the procedure for other gates such as Y, Z, H, and S single qubit gates.

Observation Table

X - Gate

$$\alpha|0\rangle + \beta|1\rangle \;—\boxed{X}—\; \alpha|1\rangle + \beta|0\rangle$$

X - Gate				
Input	**Output**	**Validation (True/False)**	**θ**	**ϕ**
$\|0\rangle$	$\|1\rangle$			
$\|1\rangle$	$\|0\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle + \frac{1}{\sqrt{2}}\|1\rangle$ (+)	$\frac{1}{\sqrt{2}}\|1\rangle + \frac{1}{\sqrt{2}}\|0\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle - \frac{1}{\sqrt{2}}\|1\rangle$ (-)	$\frac{1}{\sqrt{2}}\|1\rangle - \frac{1}{\sqrt{2}}\|0\rangle$			

Y – Gate

$$\alpha|0\rangle + \beta|1\rangle \;—\boxed{Y}—\; i\alpha|1\rangle - i\beta|0\rangle$$

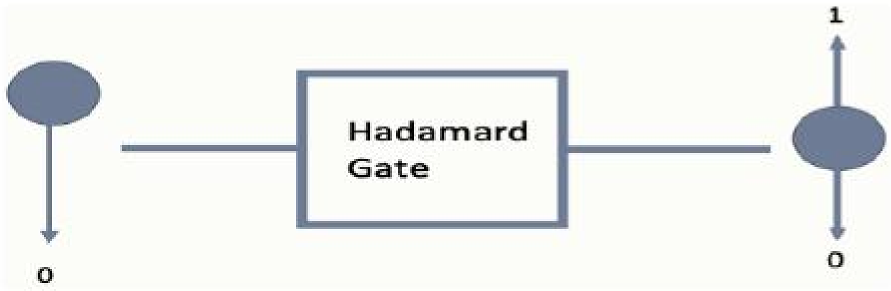
Y - Gate				
Input	**Output**	**Validation (True/False)**	**θ**	**ϕ**
$\|0\rangle$	$\mathrm{i}\|1\rangle$			
$\|1\rangle$	$-\mathrm{i}\|0\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle + \frac{1}{\sqrt{2}}\|1\rangle$ (+)	$\frac{\mathrm{i}}{\sqrt{2}}\|1\rangle - \frac{\mathrm{i}}{\sqrt{2}}\|0\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle - \frac{1}{\sqrt{2}}\|1\rangle$ (-)	$\frac{\mathrm{i}}{\sqrt{2}}\|1\rangle + \frac{\mathrm{i}}{\sqrt{2}}\|0\rangle$			

Z – Gate

$$\alpha|0\rangle + \beta|1\rangle \;—\boxed{Z}—\; \alpha|0\rangle - \beta|1\rangle$$

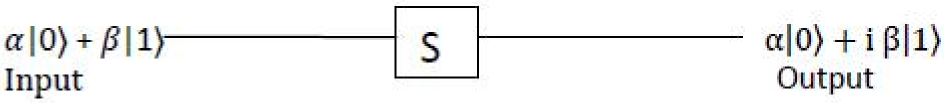
Z - Gate				
Input	**Output**	**Validation (True/False)**	**θ**	**ϕ**
$\|0\rangle$	$\|0\rangle$			
$\|1\rangle$	$-\|1\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle + \frac{1}{\sqrt{2}}\|1\rangle$ (+)	$\frac{1}{\sqrt{2}}\|0\rangle - \frac{1}{\sqrt{2}}\|1\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle - \frac{1}{\sqrt{2}}\|1\rangle$ (-)	$\frac{1}{\sqrt{2}}\|0\rangle + \frac{1}{\sqrt{2}}\|1\rangle$			

H – Gate



H - Gate				
Input	**Output**	**Validation (True/False)**	**θ**	**ϕ**
$\|0\rangle$	$\frac{1}{\sqrt{2}}\|0\rangle + \frac{1}{\sqrt{2}}\|1\rangle$			
$\|1\rangle$	$\frac{1}{\sqrt{2}}\|0\rangle - \frac{1}{\sqrt{2}}\|1\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle + \frac{1}{\sqrt{2}}\|1\rangle$ (+)	$\|0\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle - \frac{1}{\sqrt{2}}\|1\rangle$ (-)	$\|1\rangle$			

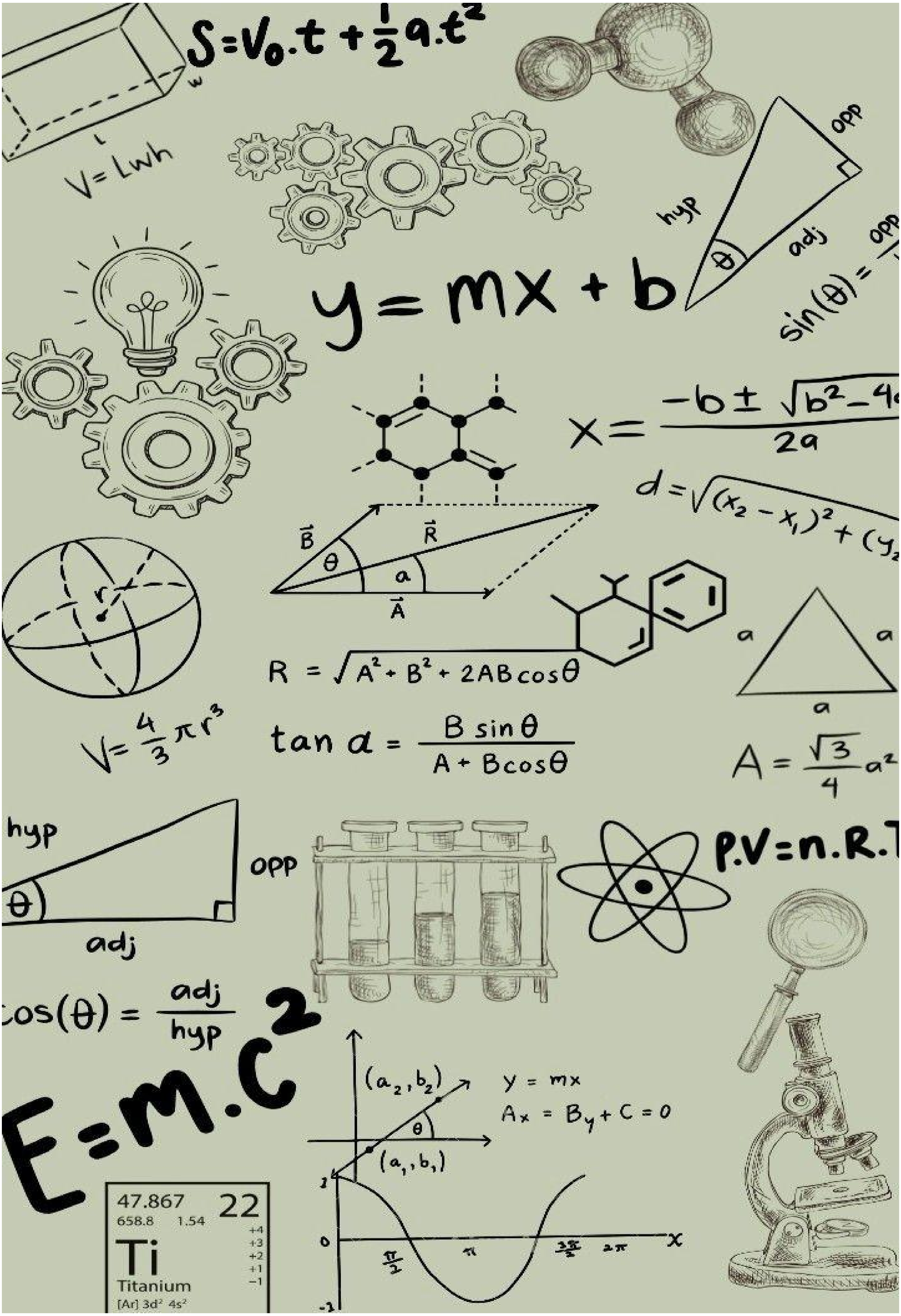
S Gate



S - Gate				
Input	**Output**	**Validation (True/False)**	**θ**	**ϕ**
$\|0\rangle$	$\|0\rangle$			
$\|1\rangle$	$i\|1\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle + \frac{1}{\sqrt{2}}\|1\rangle$ (+)	$\frac{1}{\sqrt{2}}\|0\rangle + \frac{i}{\sqrt{2}}\|1\rangle$			
$\frac{1}{\sqrt{2}}\|0\rangle - \frac{1}{\sqrt{2}}\|1\rangle$ (-)	$\frac{1}{\sqrt{2}}\|0\rangle - \frac{i}{\sqrt{2}}\|1\rangle$			

Result:

The working of quantum gates through simulation are verified and validated.



S=V0.t+1/2a.t2
V=Lwh
y=mx+b
hyp
opp
adj
sin(θ)=opp
x=-b± √b2-4
2a
d=√(x2-x1)2+(y2
B
R
θ
a
A
R = √A2+B2+2ABcosθ
a
V=4/3πr3
tan α = B sinθ / A+Bcosθ
A=√3/4a2
hyp
opp
θ
adj
P.V=n.R.T
cos(θ) = adj/hyp
E=M.C2
(a2,b2)
y = mx
Ax = By+C=0
θ
(a1,b1)
0
π/2
π
3π/2
2π
X
47.867
22
658.8
1.54
Ti
Titanium
+4
+3
+2
+1
−1
[Ar] 3d2 4s2